

Uncalibrated camera approach for vehicle speed estimation using detection and tracking

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Research Article

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Abstract

The availability of cheap and high-quality digital cameras has fostered their deployment on the roads for traffic monitoring as part of Intelligent Transport System. Implementing accurate vehicle speed estimation solution exploiting traffic surveillance footage from such widespread camera represents a credible and cost-effective alternative to traditional methods. In this work, we present an uncalibrated camera approach for vehicle speed estimation based on detection and tracking using YoloV8 and ByteTrack. The proposed method considers an average speed over a given distance by means of intrusion lines and the implementation of vector product to monitor the arrival of the traveling vehicle at expected reference locations. The experimental results demonstrated average Root Mean Squared Error and Mean Absolute Error of respectively 1.78 km/h and 1.41 km/h for vehicles traveling in the range 10 km/h – 60 km/h. These are well below the legal tolerance values for actual speed estimation systems and demonstrate the potentiality of the presented method that paves the way toward future improvement for deployment as a simple and robust solution of vehicle speed estimation.

1 Introduction

The historical need for traffic management system to strengthen speed Must often, those solutions rely on non-intrusive sensors such as Radio Detection and Ranging (RADAR) or Laser Infrared Detection and Ranging (LIDAR). However, there is a consensus concerning the expensive cost for acquiring those sensors and also for their frequent maintenance needs[1]. In addition, the RADAR based technologies are subject of errors in estimating the speed of vehicles due to cosine error, shadowing and frequency interferences such that those solution are not sufficient in today scenario [2]. In the meanwhile, Intelligent Transport System (ITS) emerged aiming to efficiently handle modern traffic multiple problems including safety, congestion, pollution and mobility. In this context, the availability of cheap and high-quality digital cameras has fostered their deployment on the roads as important component of ITS for traffic monitoring[3]. Furthermore, the recent progress in deep learning and enable developing high precision models for object detection[4], [5], [6] and tracking[7], [8] even in difficult weather condition and low quality images[9], [10]. This sets the conditions for applying object detection and image processing to road traffic footages to extract important output for ITS including vehicle speed. Implementing accurate vehicle speed estimation solution exploiting traffic surveillance video from wide road spread camera represents a credible alternative to traditional methods. In the recent years, many contributions have presented speed estimation methods based on vehicle detection and tracking combined most of with relied on camera calibration to relate image geometry to real world for converting estimated speed from pixels per hour into standard speed unit such kilometer per hour (km/h)[11]. However, whatever the method of calibration, the real-world distance error measurement from metric extracted from the image for either monocular or stereo camera is given by the $Z_{err} = \frac{1}{f_x} Z^2 p_{err} - \frac{1}{f_x} \Delta X$. Where Z is the camera to object distance, f_x the camera focal length in x axis, ΔX and p_{err} respectively the object dimension and pixel detection error for monocular camera or distance between the cameras and disparity error for stereo camera setup. Therefore, even considering the perfect calibration of intrinsic parameters of the camera either monocular or stereo an ideal scenario of zero localization Z_{err} increases quadratically with camera to object distance Z . On the other hand, it is inversely proportional to object dimension or distance between the cameras. Hence [11] concluded on three possible options to minimize the error of object distances which a key for accurate estimation of the vehicle speed, the use of telescopic lenses which have high focal length and large camera distance in stereo systems or large possible object of know dimension in monocular camera or avoid tacking measurement on vehicles that are fare from the camera. These recommendations are hard to fulfill.

Alternatively, we propose an uncalibrated camera approach for accurate vehicle speed estimation using detection and tracking based on YoloV8[12] and ByteTrack[8]. The proposed method is simple and consists of using vector product to monitor the arrival of vehicles a multiple reference lines of known real inter distances such that the subsequent speed estimation is straightforward using intrinsic video parameters. We achieve Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) of respectively 1.78 km/h and 1.41 km/h as compared to GPS ground truth. In the remaining of this paper, we present in section 2 a brief review of related work to speed detection which involves detection, tracking and distance estimation. In section 3, we detailed the proposed methodology and setup for speed estimation. In section 4 we present the results and discussion and conclude the paper in section 6.

2 Related works

2.1 Vehicle detection and tracking

The implementation of vehicle speed estimation required detection and tracking of vehicles or their representative features in all video frames. Since most traffic monitoring systems involved static camera cameras, majority of vehicle detection in the context of speed estimation are implemented by subtracting static background and identify vehicles as the foreground pixel blobs[13], [14]. The accuracy of Background subtraction based detection process is not optimal due to artefact such as object's ground shadow linked to the pixel blobs in lightning condition despite the preprocessing methods that are often conducted to optimize the extraction of region of interest [2]. Recent progress in convolutional neural network applied to object detection have enabled developing deep learning models achieving impressive detection accuracy such as Region-based Convolutional Neural Networks (RCNN)[15], You look Only Once (YOLO)[5], single shot multibox detector[4] and RetinaNet[16]. Two approaches dominate the currently trend in object detection field namely one stage detection and two stages detection. Two stage detectors such as RCNN and its variants[6], [15] include a first regions proposal generator stage using either selective method or a region proposal network. The second stage is applied to the proposed region of interest from the first stage for classification and bounding box regression. Given the clear separation of these two stages that can even be trained and inference separately, two stage object detectors usually achieve better detection as compared to one stage while been however slower. First introduce in 2016 one stage detectors YOLO[5] and single-shot detector[4], treat object detection as a regression problem where the object classes and the bounding box coordinates are predicted directly. This complete inline single stage process enables important gain of processing time. Yolo achieved original 45 frame rate processing time on a Titan X GPU that was later improved by subsequent versions which yet compete with improved two state faster RCNN detector[17] while providing real time object detection on GPU. This makes YOLO models suitable for implementing real time vehicle speed detection.

Following the detection with corresponding bounding vehicle boxes, tracking is of paramount importance to associate the detected vehicles across consecutive frames and access their individual trajectory which enables estimating their traveling distance. Many contributions applied optical flow to exploit the pattern of apparent motion of a set of vehicle features for tracking across consecutive frames [11], [18]. However, multi-object tracking is currently dominated by tracking-by-detection method which has shown to be the most effective paradigm as it provides the possibility to benefit from the efficiency of detectors to leverage the tracking performances. The general approach of multi-object tracking involves two steps. The detection to locate objects of interest in the frame and tracking to relate the detected objects to existing tracklets or create new tracklets [7]. Two strategies prevail consisting of either joint detection and embedding or separate detection and embedding. In the former case, both detection and embedding models are combined into a single deep learning network that output a single inference containing both detection results and appearance embedding features. In the latter, a first module detects targeted object bounding boxes and a second module extracts feature embedding of detected bounding boxes and proceeds with the association of each to existing trajectories or create a new.

2.2 Distance estimation

The important challenge in vision-based speed estimation algorithm is neither the extraction of metric from video frames. Instead, the accurate mapping of the extracted pixel distance metrics to real world distance to relate the vehicle speed to human understandable unit. The approaches for distance estimation in the context of speed detection can be globally classified in two categories based on the setup tool used. These include stereo camera-based approaches and Monocular-based approaches[11]. In stereo camera-based approaches each stereo pair images are cross correlated to find matching areas and construct a depth map of the scene using the apparent motion of objects between pairs of stereo images (disparity). Hence, exploiting this map provides the possibility of accessing the 3D real world coordinates of any point in the image. Therefore, the relative distance between different positions of object in the scene can be obtained from depth map as such the vehicle's travelled distance[20], [21]. Although, stereo cameras approach relies on strong calibration requirements, it does not immune them from real world distance estimation error as discussed before. Moreover, the fact that legacy traffic surveillance cameras are monocular in its large majority[11] explains the limited number of proposals using stereo cameras for

vehicle speed estimation. Using monocular camera system, a variety of approaches exist for distance estimation. Methods employing intrusion lines enable using straightforward speed estimation due to prior knowledge of distance intervals between lines particularly as traveling time is obtained from video rate or frame per second and (FPS) and number of scroll frames[3], [22]. However, this method is susceptible to error if the determination of the starting and ending location is the conducted with deep care. [23] presented a method employing intrusion lines of known inter distance to measure the movement pattern vector which further enables modeling a probability density function of a vehicle's speed and improves the speed estimate. To introduce inline distance estimation camera calibration homography method is often employed to compute a linear transformation of the 3D road scene to 2D camera image[1], [24], [25] to determine a scale factor for conversion from pixel to metric to standard distance unit based on prior knowledge of road portion. Alternative to augmented lines or homography methods for distance estimation, other methods were proposed including those based on prior knowledge of vehicles dimensions[22] or license plate's dimensions[26] to scale pixel dimension to real world dimension. Speed estimation based on methods requiring scale factor between image and real world often lack robustness due to speed fluctuation. Different mathematical formulations were also proposed[27] based histogram projection to represent vehicle motions as a group of key bins and a testing function enables selecting the corresponding vehicle speed. [28] proposed a formulation exploiting the geometry of single camera imaging system and exploited the solid angle of the real-world license plate and its corresponding in the image plane to estimate the horizontal distance from the license plate to the camera. Further, two different instances of such distance enable deducing the speed of the vehicle. Though this method is independent to the road geometry, it requires camera calibration with deeper than usual understanding of camera image formation to extract required parameters. In addition, prior knowledge of the license plate dimension is a real bottleneck for the implementation of this methods since in different countries many dimensions of license plate are most often homologated.

3 Methodology

The method we proposed consists of defining augmented reference lines perpendicular to the road plane with known real distances. The vehicle's relative position to the reference lines is continuously tracked using a vector product and bounding box coordinates of obtained from tracking. The Fig. 1 (a) presents the principle of the process which aims at determining the exact moment at which the vehicle reaches the desired reference line. Using the based coordinate system of the image (0, x, y), and without loss of generality in the context, let consider the mid bottom bounding box side as the position $\vec{M}$ of the vehicle and $\vec{A}$ $\vec{B}$ the vector belonging to a reference line (in the case the first line in Fig. 1 (a)). The vector product between the vehicle's position and the reference line ($\vec{A}$ $\vec{B}$) is defined as $\vec{A} \times \vec{M} \times \vec{B}$. When the vehicle reaches the line ($\vec{A}$ $\vec{B}$) both $\vec{A} \times \vec{M}$ and $\vec{A} \times \vec{B}$ are collinear and $\vec{A} \times \vec{M} \times \vec{B} = 0$. Once the vehicle crosses the line, the vector product changes sign in agreement with its properties. This enables to introduce the quantity Π given in Eq. 1. which is the reformulation of the vector product with the coordinate of the involved points.

$$\Pi = y - y_{\vec{A}} - \frac{(\vec{x} - \vec{x}_{\vec{A}})(\vec{y}_{\vec{B}} - \vec{y}_{\vec{A}})}{\vec{x}_{\vec{B}} - \vec{x}_{\vec{A}}}$$

1

Therefore, as the vehicle remains before the line ($\vec{A}$ $\vec{B}$), $\Pi < 0$. As the vehicle crosses the line (AB), Π changes sign in agreement with vector product properties. The Fig. 1 (b) presents the evolution of Π as refer to the first and third reference line of Fig. 1 (a). The annotation in Fig. 1 (b) highlight the moments (frames) at which Π becomes nil before to increase with positive sign. When a given tracked vehicle reaches a reference an intrusion line, the frame index and its track id are recorded. As such, starting from the second intrusion line, the vehicle's speed estimation is straightforward without any further processing using Eq. 2.

$$V(\vec{k} \vec{m} / \vec{h}) = \frac{3.6 \bullet \vec{F} \vec{P} \vec{S}}{\vec{N}} \bullet \vec{d}$$

Where $\varvec{N}$ is the number of scrolled frames between consecutive intrusion lines, d the known distance between the corresponding intrusion lines and $F/P/S$ the frame rate per second. The coefficient 3.6 enables the conversion from meter per second to kilometer per second. This approach relies on prior knowledge of distance of elements on the road. Often, the marks on the road are enough to have access to such a distance given that they have homologated length in many countries. The principle of our algorithm is that each time a vehicle reaches the first reference line the first time, it tracks id if saved together with relevant parameters such as the frame index. When the vehicle reaches the following reference lines, the algorithm checks that it first crossed the previous lines with relevant parameters available, and process with speed estimation. Starting from third augmented line, it is possible to combines the speed different line intervals and estimate the average speed. This may help reducing the eventual error that may occur when estimating a single speed.

Contrary to camera calibration-based distance estimation methods where the speed estimation accuracy depends on the precision of the calibration procedure, the speed estimation accuracy of our methods depends on the accuracy of tracking and the localization of the vehicle at the expected augmented line locations. In fact, the only condition of positive $\varvec{II}$ value is not enough to guarantee the detection at the exact reference lines location. Indeed, misdetection and tracking on few frames around the reference lines may introduce non negligible error on number of frames and thus the speed accuracy. To attenuate misdetection and tracking our system relies on the best balance between the state-of-the-art detection and tracking model and processing speed. Hence, system is based on YoloV8 for detection and ByteTrack[8]. Based on its accuracy, YoloV8[12] is a state-of-the-art model developed that leverage the advantages of previous YOLO version and introduces new feature and improvements to further boss the performances.

4 Experimental results

The experiments were conducted in the Gyrovia trail which is an experimentation and demonstration road of approximately 850 meters with two lanes dedicated to research, training, promotion, and development of new technologies in the domain of transport an autonomous vehicle. Our experimental setup consisted of an IP camera of 25 FPS rate connected to a computer equipped with GeForce TRX 20270 graphic carte together with our system deployed for real time speed estimation. The camera was installed on roadside at a high of approximately 2 m for better view of the scene. At this stage, the experiment consisted of driving cars at a speed from 10 to 60 km/h through the scene and estimate its speed in real time. Prior to speed estimation, we conducted a verification of conformity of known parameters such as locations of detection and particularly traveled distance with the known values at expected detection locations. This is important because the behavior of the system in the image plane may differ from the expectation.

The Fig. 2 shows the detection of a vehicle a two distinct location lines. In Fig. 2 (a), the first detection is conducted and the first reference line. This specific experiment was conducted with 4 intrusion lines as can be observed in Fig. 1 (a). The distances were $d = [24.65, 13.29, 13.29]$ (m) between lines corresponding respectively to line 1 and 2, line 2 and 3 and line 3 and 4. The next detection could be conducted and any of the 3 other lines. The Fig. 1 (b) shows the detection at line 3. One observes that the module successfully captures the arrival of the vehicles at the exact reference lines. This ability to accurately monitor the arrival of vehicles position at the expected reference locations validates our approach for speed estimation and augurs its accuracy given that from there on, the speed estimation simply consists of counting scrolled frames and applied the Eq. 2 to estimate the speed. In Fig. 3 we show a line plot of the speed estimation compile from two different experimental sessions with 89 and 70 instances respectively. We implemented in the application a report file such that the speed prediction at any run is conceived automatically. The footages of different experiment runs were recorded such that we could reproduce the experiment afterward. One observes that as expected, we have satisfactory corresponding between base line speeds and estimations.

We computed different statistics to gain better understanding of the magnitude of speed estimation. The Mean Absolute Error (MAE) takes the sum of absolute difference between individual predicted and true instance and divides it by the total number of instances. Given the linear expression of MAE, it gives equal importance to each instance error. The Root Mean Squared

Error (RMSE) computes the squared root of the Mean Squared Error that is the average difference between predicted and real speed. By squaring the difference between true and predicted values, RMSE gives relatively high weight to large error. A combination of both metrics enables a good understanding of the error magnitude between estimated speed and corresponding GPS ground truth. For experiments presented in Fig. 3, we observed MAE and RMSE of respectively 1.82 km/h and 1.38 km/h for experiment in Fig. 3 (a) and 1.75 km/h and 1.3 km/h for experiment in Fig. 3 (b). Considering the simplicity of our approach these values of RMSE are of great satisfaction given that they are well inferior to the legal error authorized for speed detection equipment which is for example in France 5 km/h for speed inferior to 100 km/h and 5% of the estimated speed above 100 km/h. We also found that the estimation of speed in the range of 20 km/h and below induced larger errors as compared to order ranges. In table I, we report the RMSE for the speed estimation presented in Fig. 3. One observed a noticeable difference of RMSE values when conducting separating statistics for instances with speed below and above 20 km/h. We believe this was related either to the difficulty to stabilize at low speed or the difficulty of the GPS provides a precise ground truth.

Table 1
RMSE for two experimental sessions and for speed above and below 20 km/h

Exp. session	RMSE	RMSE	
		> 20 km/h	<= 20 km/h
1	1.82 km/h	1.23 km/h	3.90 km/h
2	1.75 km/h	1.60 km/h	2.1 km/h
Average	1.78 km/h	1.41 km/h	3.0 km/h

Based on the configuration with multiple intrusion lines (see Fig. 1 (a)), we present in Table 2 the RMSE and MAE together with the corresponding average of experiments conducted while estimating the speed at different augmented line intervals. The statistics concerned 49 instances of cars running through the scene. One observes the average statistics of 1.90 km/h and 1.47 km/h for RMSE and MAE respectively remains in similar ranges as the results of Table 1.

Table 2
RMSE and MAE and corresponding average of speeds estimated at different augmented lines.

	RMSE	MAE
REF. 12	1.93 km/h	1.50 km/h
REF. 13	1.87 km/h	1.49 km/h
REF. 14	1.91 km/h	1.44 km/h
Average	1.90 km/h	1.47 km/h

Furthermore, both RMSE and MAE of each set of speeds estimated at given reference lines do not show a noticeable difference with the others. This observation further demonstrates the robustness of the approach.

5 Discussion

Conceptually, the accuracy of the speed estimation is not surprising. The vehicle detection at two intrusion lines with known exact distance, number of flow frames enables straightforward calculation of the speed. Therefore, the accuracy of the method relies mostly on the ability to conduct the detection at the expected locations. In fact, depending on the camera position and orientation, the correspondence in pixel of the vehicle's position with the intrusion lines is not always compatible with the reality. This may introduce a non-negligible mismatch between the traveled distance and the real distance between lines and thus affect the accuracy of the estimated speed. Fortunately, we observed that the behavior of detection is systematic such

that in a situation where eventual mismatch appears between detection location and expected location, is possible to inspect the location of detection and adjust the distance to avoid discrepancy. The Fig. 2 also provides an illustration of such visual inspection. Further, in condition of low image quality, some tracklets can be lost during a few frames before being later recovered or redetected with a new track id. If this happens across an intrusion line, the speed for the corresponding tracklet could not be estimated due to lack of information such as frame index. In such condition, multiple intrusion lines (> 2) may be important for estimating the speed at other lines at which the mentioned vehicle has been detected.

6 Conclusion

We presented a workflow for vehicle detection, tracking, and speed estimation using an uncalibrated camera approach. The detection and tracking were conducted using YOLOv8 and ByteTrack. We implemented speed estimation by means of intrusion lines with prior knowledge of the corresponding distances between pairs of lines. The product vector was exploited to continually monitor the arrival of the vehicle at the expected reference lines. Each time a new vehicle with a new tracked id reaches an intrusion line, its corresponding parameter is recorded one for speed estimation. Our experimental results demonstrated the best RMSE and MAE of respectively 1.78 km and 1.41 km/h for vehicles traveling in the range 10 km/h – 60 km/h. Thanks to the performances of the YOLOv8 and ByteTrack models, we observed no noticeable issue with the detection and tracking accuracy during all experiments conducted. The proposed approach considers an average speed over a defined distance and therefore is more robust to fluctuating speed unlike other methods based on a scale factor that may necessitate taking multiple speed measurements throughout monitoring to account for frequent changes in vehicle acceleration.

Abbreviations

RMSE
Root Mean Square Error
MAE
Mean Absolute Error
YOLO
You look Only Once
GPS
Global Positioning System
FPS
Frame Per Second
RADAR
Radio Detection and Ranging
LIDAR
Laser Infrared Detection and Ranging
ITS
Intelligent Transport System
GPU
Graphics Processing Unit

Declarations

Consent for publication

Study participants provided written consent to publish their identifying images.

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Author Contribution

D.M. and U.B. conceived the presented idea and carried out the experiment. D.M. developed the theory and performed the analysis and wrote the algorithms. Both authors discussed the results. D. M. wrote the final manuscript.

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Availability of data and material

The video generated during the current study are not available publicly due to legal constraints but are available for private access from the corresponding author on reasonable request.

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Footnotes

1. The Gyrovia Trail (transalley.com)